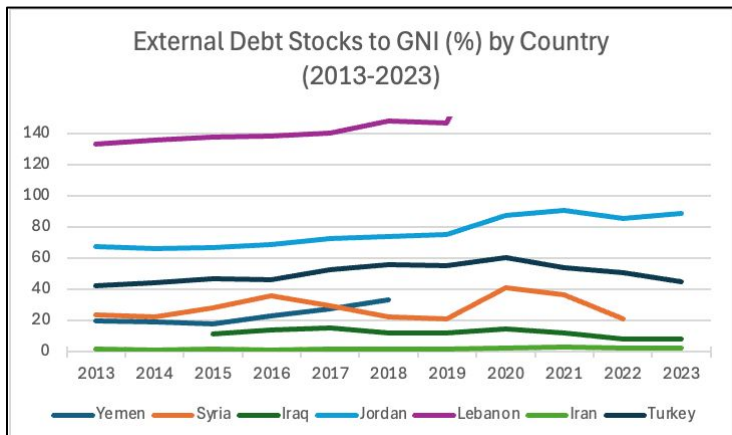


Measuring the Relationship between External Debt and Life Expectancy at Birth for Middle Eastern Countries



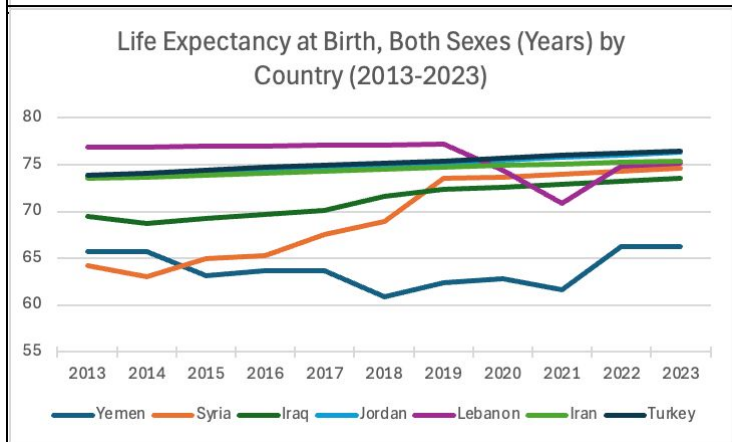
X-Axis: Year

Y-Axis: Debt as a percentage of Gross National Income

Color: Country

Data: World Bank's International Debt Statistics

(<https://www.worldbank.org/en/programs/debt-statistics/ids>)



X-Axis: Year

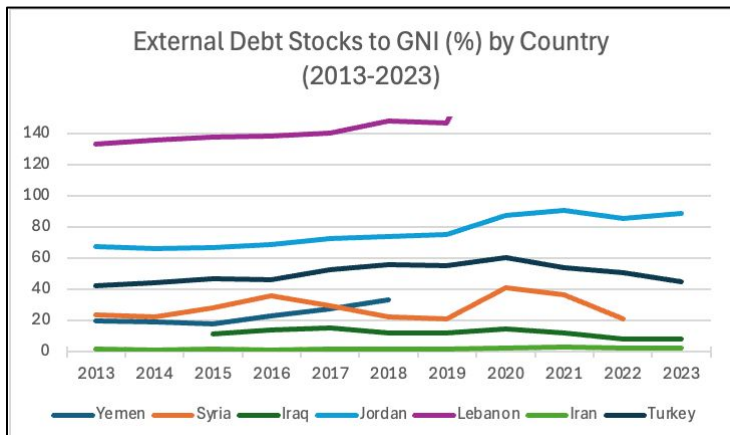
Y-Axis: Life Expectancy at Birth for Both Sexes

Color: Country

Data: US Census Bureau's International Database

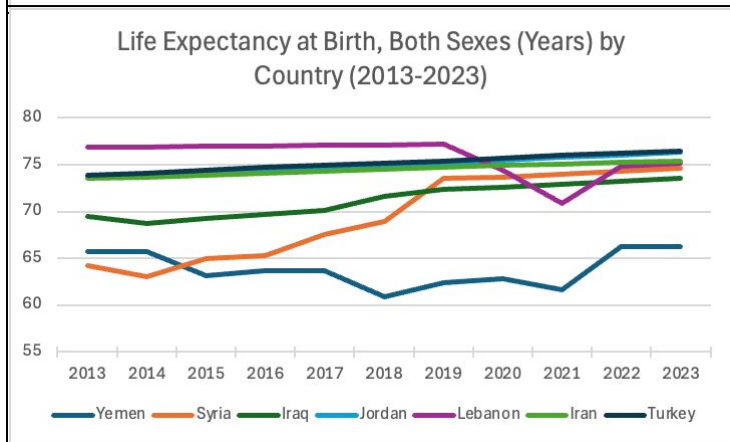
(https://www.census.gov/data-tools/demo/idb/#/table?dashboard_page=country&COUNTRY_YR_ANIM=2025&CCODE_SINGLE)

Measuring the Relationship between External Debt and Life Expectancy at Birth for Middle Eastern Countries



Key Observations:

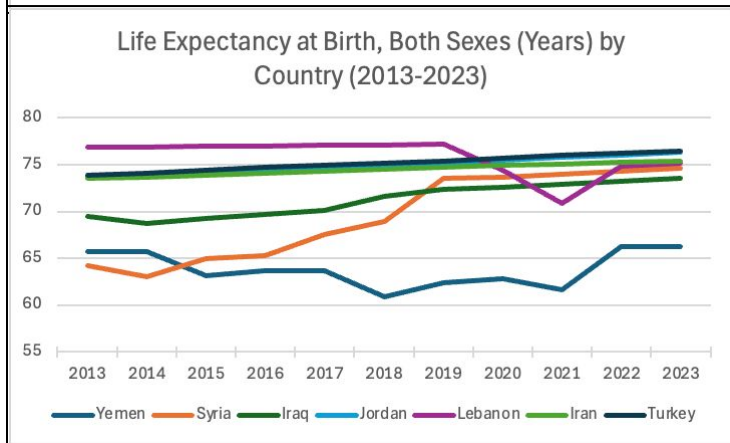
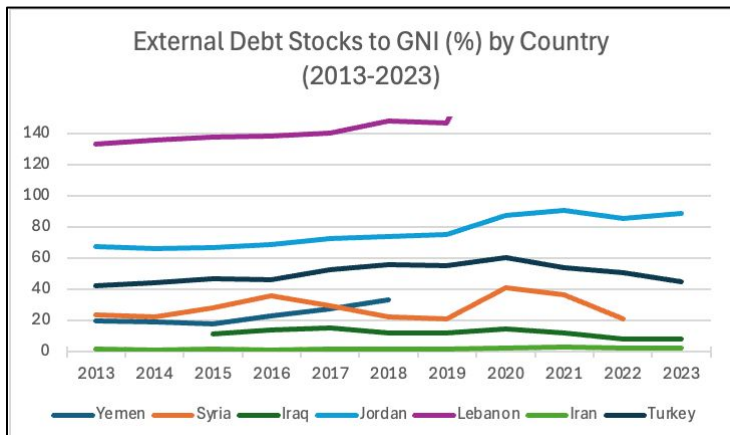
- Lebanon's Debt/GNI exceeds 300% (315% in 2021, 332.8% in 2022); Lebanon also sees the largest change across this time horizon with its beginning value at 133.1% in 2013 before peaking at 332.8% in 2022
- Iran's Debt/GNI is fairly flat only near 0 ranging between 1.3% in 2014 to 2.7% in 2021
- Data is limited on Iraq pre-2015 and Yemen post-2018
- Across this time horizon, debt/GNI levels increased in Jordan, fluctuated to reach near 2013 levels in 2023 again for Turkey and Syria



Key Observations:

- Life expectancy at birth increased fairly linearly for Jordan, Turkey, Iran
- Turkey's life expectancy ended as the highest of the Middle East in 2023
- Yemen's life expectancy dropped after 2014 to eventually reach a low in 2018 (60.9 years), but bounced back to 66.2 by 2022
- Syria saw the greatest increase across this time horizon, beginning at 64.2 in 2013 and ending at 74.6 in 2023 (+16% increase)
- Lebanon's life expectancy began as the highest in the region at 76.9 in 2013, but dropped to a low of 70.9 in 2021, before slightly recovering back to 75.2 in 2023 (but no longer the greatest)

Measuring the Relationship between External Debt and Life Expectancy at Birth for Middle Eastern Countries



Initial Hypothesis: Countries with a higher debt/GNI percentage would have a lower life expectancy at birth.

Supplemental Findings:

- As Lebanon's debt/GNI increased +114% between 2019 - 2021 (146.7 to 315), its life expectancy at birth decreased -8.2% (77.2 to 70.9)
- Syria saw a -41.4% decrease in its debt/GNI metric between 2016 and 2019 (36 to 21.1), while its life expectancy at birth increased +12.6% during this time (65.3 to 73.5)
- Iraq saw a -31% decrease in its debt/GNI number between 2015 to 2020 (11.6 to 8.0), while its life expectancy at birth exhibited a +5.6% increase during this time (69.3 to 73.2)

Conclusion: Some countries (3/7) align with the hypothesized relationship, but since 4/7 did not, there is not sufficient proof to confirm the association between debt/GNI levels and life expectancy at birth.